

Project Brief - Customer Personality Analysis

Group 4

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Project Summary

Objective: To analyze the effectiveness of discounts on customer purchasing behavior and develop strategies to enhance sales and customer engagement.

Interest Groups: Marketing teams, sales departments, and senior management would be keenly interested in the outcomes to refine discounting strategies, enhance customer engagement, and boost revenue.

Furthering Work: This project aims to build on existing customer data analytics, enhancing understanding of how discounts affect purchasing across different channels and customer segments.

Data Utilized: Customer demographics: ID, Year of Birth, Education, Marital Status, Income; Household composition: Kidhome, Teenhome; Engagement and purchasing behavior: Recency, NumDealsPurchases, MntProducts (Wines, Fruits, etc.), Campaign Responses.

Analysis Methods: Correlation analysis, customer segmentation, and Comparative Analysis to assess promotional vs. regular purchase behaviors. Descriptive statistics to summarize data characteristics. Visualization techniques (histograms, box plots, scatter plots) to identify patterns and outliers.

Beneficiaries: The direct beneficiaries will be the marketing team for crafting precise discount strategies which can use insights to optimize discount strategies. Product teams for aligning offerings with customer preferences.

Outcome and Impact: A comprehensive EDA report highlighting key insights on discount effectiveness and customer behavior. Identification of potential opportunities for increasing sales and customer engagement through targeted discounts.

Overview of Needs

Team and Sponsor Expectations:

- Gain deep insights into customer purchasing behavior influenced by discounts.
- Understand the effectiveness of discount strategies on various customer segments.

Current Gaps and Problems:

- Limited understanding of how discounts impact customer purchasing patterns.
- Inefficiencies in targeting the right customer segments with promotional offers.
- Lack of data-driven insights to guide marketing and sales strategies.

What EDA Aims to Solve:

- Identify Patterns and Trends: Uncover hidden patterns in customer behavior that are not apparent without in-depth data analysis.
- Spot Anomalies: Identify outliers in purchasing behavior that may indicate opportunities or issues with current marketing strategies.
- Understand Customer Segments: Provide a clearer picture of how different segments react to discounts and promotions.
- Guide Resource Allocation: Help in optimizing marketing spend by focusing on the most responsive customer segments.

Expected Benefits:

- Enhanced targeting of marketing campaigns to increase effectiveness and ROI.
- Better product placement and promotional offers aligned with customer preferences and behaviors.
- Strategic insights that enable the company to adapt to customer needs more dynamically.

Project Deliverables

Agreed Project Deliverables:

A detailed document that includes statistical summaries, data visualizations, and initial findings on customer purchasing behaviors in relation to discounts. For example:

- Descriptive statistics summary for each variable.
- Visual data analysis including histograms, box plots, and scatter plots.
- Correlation analysis report between different types of purchases and discount usage.
- Customer Segmentation Analysis: Identification of distinct customer segments based on their purchasing behavior, preferences, and responsiveness to discounts.
- Insight Presentation: A presentation summarizing key insights, patterns, anomalies, and potential strategic recommendations derived from the EDA.

Data Summary

What Data Do We Have?

- Customer Demographics: Includes ID, Year of Birth, Education, Marital Status, and Income.
- Household Composition: Data on Kidhome (children in the household) and Teenhome (teenagers in household).
- Purchasing Behavior & Preferences: Amounts spent on various product categories over the last 2 years, including wines, fruits, meat, fish, sweets, and gold.
- Engagement and Promotional Data: Information on the number of deals purchases, responses to marketing campaigns (AcceptedCmp1-5, Response), and the recency of purchases.
- Shopping Channels: Data on the number of purchases made through the company's website, catalog, and in stores, as well as the number of website visits in the last month.

What Does Each Dataset Represent?

- Provides a comprehensive overview of customer profiles, their purchasing habits, responsiveness to marketing efforts, individual customer's purchases, interactions, responses to marketing efforts, and preferred shopping channels.
- Data Source: Likely from the company's CRM system, capturing information from both online transactions and in-store activities.

Data Gaps and Potential Influence on Outcomes:

- Missing Data: Specifics on customer satisfaction or feedback on purchased products are not available. External factors like market trends, and competitor information, not included
- Influence: Without customer satisfaction data, it's challenging to correlate purchasing behavior with satisfaction levels, possibly overlooking insights on customer loyalty and product improvement opportunities.

Out-of-Scope Deliverables:

- Anything requiring data not provided (like external market analysis) or the development of new software tools is out of scope.
- Customer Feedback Analysis: Due to the absence of direct customer feedback or satisfaction data in the dataset, analyzing customer sentiment is out of scope.

Proposed Methodology

Data Preparation and Cleaning: This initial phase involves cleaning the data to ensure accuracy and completeness. It includes handling missing values, correcting data anomalies, and standardizing data formats. This step ensures the data is reliable for analysis.

Exploratory Data Analysis (EDA): EDA will be conducted to understand the data's underlying structures, patterns, and anomalies. This includes visualizing distributions of key variables, identifying outliers, and understanding relationships between variables. The goal is to gain insights that will guide further analysis.

Principal Component Analysis (PCA) is a statistical technique used to reduce the dimensionality of datasets, enhancing interpretability while minimizing information loss. This method transforms a set of possibly correlated variables into a smaller set of uncorrelated variables, known as principal components.

Project Success

Impact on Sales and Engagement: The primary measure of success will be whether the analysis leads to an improvement in sales or customer engagement. This implies that the insights gained from examining the impact of discounts on customer purchasing behavior are effectively translated into actionable strategies.

Optimizing Discount Strategies: Success also means finding the right balance between offering discounts and maximizing profit. The project should provide clear guidelines on how to structure discounts to appeal to customers without eroding profit margins.

The ultimate outcome of the project should be a set of data-driven recommendations that enable the company to use discounts more strategically to drive sales and build customer relationships, thereby increasing profitability and market competitiveness. This involves not just a one-time boost in sales but fostering a sustainable model for customer engagement and pricing strategy that supports long-term business goals.

User-Stories

User Story 1: Marketing Manager - Positive Impact

- Scenario: Upon utilizing insights from the EDA report, the marketing manager is able to launch highly targeted promotional campaigns.
- Outcome: Increased campaign conversion rates and better allocation of marketing budget.

User Story 2: Customer Service Representative - Positive Impact

- Scenario: Insights into customer segments enable personalized service approaches, improving customer satisfaction.
- Outcome: Higher customer retention rates and positive feedback.

User Story 3: Product Development Team - Positive Impact

- Scenario: The EDA uncovers a strong preference for certain products among key segments, guiding product innovation.
- Outcome: Successful launch of products tailored to customer preferences, leading to increased sales.

User Story 4: Customers - Positive Impact

- Scenario: Customers benefit from promotions and products that are more closely aligned with their preferences and needs.
- Outcome: Enhanced shopping experience and perception of value, leading to greater loyalty.

User Story 5 Small Segment Customers - Negative Impact

- Scenario: Marketing efforts become so targeted that customers in smaller, less profitable segments feel overlooked.
- Outcome: Decreased satisfaction among these customers, who may feel the company no longer caters to their needs.

Project Plan and Milestones

Phase 1 - Data Preparation and Preliminary Analysis

- Success: Clean, comprehensive datasets ready for analysis.
- Involvement: Data analysts, data extraction and cleaning.
- Demo: Presentation of data quality and initial insights.

Phase 2 - Detailed Analysis and Modeling

- Success: Identification of key correlations
- Involvement: Data scientists, marketing analysts.
- Demo: Initial model results and findings report.

Phase 3 - Strategy Development and Recommendations

- Success: Comprehensive set of actionable recommendations for discount optimization.
- Involvement: Marketing team, sales department.
- Demo: Strategy presentation and discussion.

Phase 4 - Implementation (Post-Project)

- Success: Implementation of recommendations and positive impact on sales and customer engagement.
- Involvement: Marketing and sales departments, with ongoing analysis.
- Demo: Review of sales performance and adjustments to strategies as needed.

Project Roles and Responsibilities

Name	Role	Responsibilities
Siddhant Gadhe	Data Analyst	Data Preparation and Cleaning
Kshitij Kamat	Data Analyst	Exploratory Data Analysis (EDA)
Shilong Ye	Data Analyst	Principal Component Analysis (PCA):